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# SDformer: Spiking Dilated Self-Attention for Efficient Spiking Transformers in Vision Tasks

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**Abstract**—Spiking transformers integrate SNNs with Spiking Self-Attention and have demonstrated impressive performance. However, existing spiking transformers suffer from two fundamental limitations. One is that the max pooling operation in Spiking Tokenizer captures only the strongest response and fails to comprehensively preserve representative regional features. The other is that Spiking Self-Attention involves all global feature interactions, which results in substantially redundant and irrelevant computations, thus conflicting with the sparse and energy-efficient characteristics of SNNs. To address these challenges, we develop SDformer, a novel spiking transformer that incorporates Spiking Response Pooling (SPooling) and Spiking Dilated Self-Attention (SDA). For the first time, our SDA leverages a local dilated window mechanism to capture both local details and long-range dependencies without increasing computational cost. Experimental results demonstrate that our SDformer achieves state-of-the-art performance compared to existing advanced spiking transformers, and highlight the potential of SDformer to advance energy-efficient spiking models toward practical deployment in vision applications.

**Index Terms**—Spiking Neural Networks, Spiking Transformers, Spiking Dilated Self-Attention, Spiking Response Pooling, Spiking Dilated Transformer

## I. INTRODUCTION

Brain-inspired Spiking Neural Networks (SNNs) are regarded as the third generation of neural networks [1]. Compared with traditional Artificial Neural Networks (ANNs), SNNs process information via binary spikes. The spike neurons of SNNs only transmit spikes when their accumulated membrane potential exceeds the firing threshold, otherwise remaining inactive. This forms an event-driven sparse processing paradigm of SNNs and offers significant improvements in energy efficiency compared to conventional ANNs. The exceptional performance of SNNs is evident in various intelligent applications, including pattern recognition [2], target tracking [3], and robot navigation [4].

There are two primary methods to obtain SNNs, one approach involves converting pre-trained ANNs into SNNs [5], [6], but this approach demands an extensive number of timesteps. The other method directly trains SNNs using temporal-spatial information through backpropagation with gradient descent [7], [8], and this approach reduces the demand for timesteps. However, there remains a substantial inferior performance between SNNs and ANNs. Particularly in complex

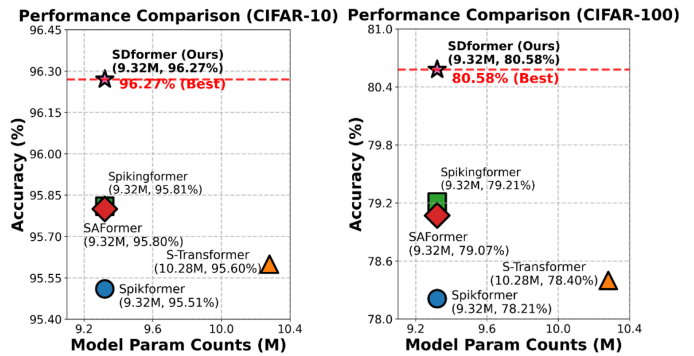


Fig. 1: Performance of SDformer and other spiking transformers regarding accuracy and model params. Results show that our proposed SDformer achieves higher accuracy compared to other models while maintaining comparable model params.

computer vision tasks, restricting the broader application and advancement of SNNs.

Researchers have made significant efforts to address the performance bottlenecks of SNNs. Since the transformer [9] has demonstrated exceptional performance across diverse domains, including natural language processing [9]–[11] and computer vision [12]–[14]. Building on the success of transformer, recent studies have increasingly explored its integration with SNNs to develop spiking transformers [16]–[22], which combine the powerful ability of transformer and also leverage the low-energy consumption characteristics of SNNs.

Spiking transformers [16]–[22] project original images into spike-form patch sequences through Spiking Tokenizer, and further extract spatial and channel features to produce high-quality representations by stacking encoders. Each encoder consists of a Spiking Self-Attention (SSA) and a MLP block. Similar to the Vanilla Self-Attention mechanism [9], the attention matrix of SSA is computed by multiplying the query matrix with the transposed key matrix. The value matrix is then used to aggregate the relevant information, where each element is weighted by the attention matrix to produce the final output. Notably, the attention matrix has natural non-negativeness, and do not need the softmax (required in Vanilla Self-Attention mechanism) to keep the attention matrix non-negative. Despite

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the advances of spiking transformers over previous SNNs, they are still constrained by two critical limitations that impede their efficiency and performance.

One limitation is that Spiking Tokenizer employs the max pooling layers to reduce the size of feature maps. However, the max pooling only retains the strongest response within each pooling window, failing to comprehensively preserve representative regional features, thus leading to significant information loss [23], [24].

The other limitation involves the SSA, whose computations of global self attention depend on dense matrix operations across all global feature interactions. Previous studies [13], [25]–[27] have shown that task-relevant areas are often locally concentrated, implying that global receptive field of SSA introduces substantial redundant and irrelevant computations, limiting its ability to effectively capture discriminative features. Several studies [25]–[27] introduce window-based self-attention to restrict attention computation within the small neighborhood windows, which effectively reduces redundant and irrelevant computations. However, the performance of window-based self-attention heavily depends on the empirically chosen window sizes. Specifically, small windows limit the capability to model long-range dependencies, whereas larger windows increase computation and memory costs. Furthermore, the fixed window size lacks flexibility in modeling multiple receptive fields.

In this work, we propose the Spiking Dilated Transformer (SDformer), a novel architecture that integrates the Spiking Response Pooling (SPooling) and the Spiking Dilated Self-Attention (SDA) mechanism. SPooling effectively exploits the complementary strength of max pooling and average pooling, enabling to capture more comprehensive information. SDA computes dilated self-attention across multiple receptive fields to efficiently capture both local details and long-range dependencies without increasing computational cost.

It is noteworthy that real-world objects have directional dependencies [28]. Our proposed SDA independently models directional features along horizontal and vertical windows to improve the ability of model for capturing spatial directional variation. Objects often exhibit multi-scale structural characteristics [28]. SDA extracts multi-scale information of the feature maps in different receptive fields to identify objects of varying scales. Our main contributions are summarized as follows:

- We propose the Spiking Dilated Self-Attention mechanism, which efficiently extracts both fine-grained local details and long-range dependencies without additional computational overhead, while modeling directional dependencies along horizontal and vertical windows.
- We propose Spiking Response Pooling to alleviate the limitations of max pooling and mitigate the loss of effective information in Spiking Tokenizer.
- Extensive experiments demonstrate that our method outperforms state-of-the-art SNNs on various static image datasets, achieving 96.27% accuracy on CIFAR-10, 80.58% on CIFAR-100 with only 4 timesteps. On neuromorphic datasets, results on CIFAR10-DVS (83.0%)

and DVS128 Gesture (98.6%) further validate the generalizability of our method.

## II. RELATED WORKS

### A. Spiking Transformers

Spiking transformers integrate SNNs with transformer architecture, attracted significant attention due to their potential for low energy consumption and high performance. Spikformer [17] introduces Spiking Self-Attention (SSA), which replaces conventional dense Query, Key, and Value matrices with sparse spike-driven activations. SSA [17] effectively eliminates energy-intensive matrix multiplications and the softmax operation, leading to substantial energy savings. Building on this, Spikformer [18] redesigns the residual connection paradigm, yielding a fully spike-driven architecture that minimizes non-spiking computations. The ConvBN-MaxPooling LIF (CML) downsampling module [19] is proposed to address the challenge of insufficient gradient propagation in spiking models. Further advancements include Spike-Driven Transformer [20], which incorporates a novel Spike-Driven Self-Attention mechanism that replaces multiplication operations entirely with masking and addition, thereby reducing computational complexity and energy consumption. The Temporal Interaction Module (TIM) [21] enhances temporal processing in SNNs by adaptively fusing historical and current information. SWformer [16] effectively learns comprehensive spatial-frequency features in a spike-driven manner by leveraging the sparse wavelet transform. SAFormer [22] designs the Spike Aggregated Self-Attention (SASA) mechanism, which significantly simplifies the computation process by calculating attention weights using only the spike matrices query and key, effectively reducing energy consumption.

Despite these advances, existing spiking transformers predominantly rely on global receptive field self-attention mechanisms, which incur substantial computational redundancy. Moreover, they generally lack explicit mechanisms for modeling multi-scale receptive fields, limiting their effectiveness in complex vision tasks.

## III. METHODS

In this section, we introduce the overall architecture of SDformer (Section III-A), a novel downsampling mechanism Spiking Response Pooling (Section III-B), and the Spiking Dilated Self-Attention (Section III-C).

### A. Overall Architecture of SDformer

As illustrated in Fig. 2, our proposed SDformer contains a Spiking Tokenizer (ST), a stack of  $L$  encoder blocks, and a linear classification head. Given a 2D image sequence  $I \in \mathbb{R}^{T \times C \times H \times W}$ , where  $T$ ,  $C$ ,  $H$ , and  $W$  denote timestep (for static image datasets,  $T$  is the number of repetitions), channel, height and width of image sequence, respectively. We employ the Spiking Tokenizer to project the image sequence into the  $D$ -dimensional spike patch sequence  $U_0$ .

$$U_0 = \text{ST}(I), I \in \mathbb{R}^{T \times C \times H \times W}, U_0 \in \mathbb{R}^{T \times D \times H' \times W'} \quad (1)$$

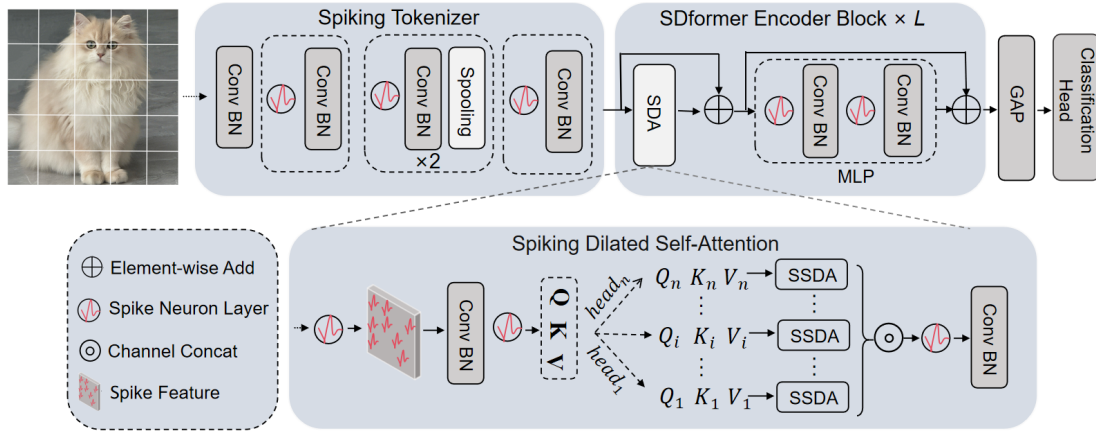


Fig. 2: The overall architecture of our proposed SDformer, which integrates a Spiking Tokenizer module, multiple SDformer encoder blocks, and a classification layer. Compared with typical transformer-based SNNs (e.g., Spikingformer [18]), SDformer replaces the conventional SSA with our proposed SDA, and replaces the maxpooling with Spooling. Note that ConvBN denotes a two-dimensional convolution layer (stride-1,  $3 \times 3$  kernel size) followed by batch normalization.

Here,  $H'$  and  $W'$  represent the height and width of  $U_0$ , respectively. For static image datasets, the values of  $H'$  and  $W'$  are obtained through two Spooling operations (stride-2,  $3 \times 3$  kernel size) in the Spiking Tokenizer.  $D$  represents the channel dimension of  $U_0$  (according to [16]–[20], [22],  $D$  is set to 384 for static datasets and 256 for neuromorphic datasets). Then we pass  $U_0$  to a stack of  $L$  encoder blocks. Each encoder block consists of a SDA (see Section III-C) and a MLP block [18]. Residual connection is employed for membrane potential in both the SDA and the MLP. Signal flow is formulated as:

$$U_l = \text{SDA}(U_{l-1}) + U_{l-1}, U_l \in \mathbb{R}^{T \times D \times H' \times W'}, l = 1 \dots L \quad (2)$$

$$S_l = \text{MLP}(U_l) + U_l, S_l \in \mathbb{R}^{T \times D \times H' \times W'}, l = 1 \dots L \quad (3)$$

Here,  $U_l$  and  $S_l$  represent the outputs of the  $l$ -th layer of the SDA and the encoder block, respectively. Finally, a global average-pooling (GAP) is utilized on the output of the last encoder block to yield  $D$ -dimensional feature, which is sent to the fully-connected-layer classification head (CH) to output the prediction  $Y$ .

$$Y = \text{CH}(\text{GAP}(S_L)) \quad (4)$$

### B. Spiking Response Pooling

The pooling layers are widely used for downsampling feature maps, with common approaches including max pooling and average pooling. To effectively leverage the complementary strength of these two methods, we propose Spiking Response Pooling (SPooling), whose implementation consists of two core steps as follows.

Specifically, SPooling first introduces a threshold  $\lambda$  to evaluate the average response level within each pooling window. Let  $i$  and  $j$  denote the spatial coordinates of elements in the

feature map. the filtering process for the average response is formulated as:

$$x'_{\text{avg}}(i, j) = \begin{cases} x_{\text{avg}}(i, j), & x_{\text{avg}}(i, j) \geq \lambda \\ 0, & x_{\text{avg}}(i, j) < \lambda \end{cases} \quad (5)$$

Here,  $x_{\text{avg}}(i, j)$  represents the average pooling result at position  $(i, j)$ , and  $x'_{\text{avg}}(i, j)$  denotes the filtered average response after thresholding. As shown in Eq. 5, when  $x_{\text{avg}}(i, j)$  exceeds the threshold  $\lambda$ , it is considered to contain meaningful features and retained. In contrast, the values of  $x_{\text{avg}}(i, j)$  below the threshold are filtered out.

Second, to fuse the advantages of max pooling and the filtered  $x'_{\text{avg}}(i, j)$ , we introduce a learnable gating parameter  $\theta$  to dynamically regulate the contribution of  $x'_{\text{avg}}(i, j)$ . The final output of SPooling at spatial position  $(i, j)$  is computed as:

$$x(i, j) = x_{\text{max}}(i, j) + \sigma(\theta) \cdot x'_{\text{avg}}(i, j) \quad (6)$$

Here,  $x_{\text{max}}(i, j)$  is the max pooling result at position  $(i, j)$ ,  $\theta$  is the learnable gating parameter, and  $\sigma(\cdot)$  is the sigmoid function, which constrains  $\theta$  within the range  $(0, 1)$  to ensure stable gradient propagation.

### C. Spiking Dilated Self-Attention

Given the spike-form sequence  $X \in \{0, 1\}^{T \times D \times H' \times W'}$ , where  $T$ ,  $D$ ,  $H'$ , and  $W'$  represent the timestep, channel, height and width of the  $X$ , respectively.  $X$  is mapped to generate spike-form query (Q), key (K), and value (V) matrices. This mapping process is formulated as:

$$Q = \text{SN}_Q(\text{BN}_Q(\text{Conv}_Q(X))) \quad (7)$$

$$K = \text{SN}_K(\text{BN}_K(\text{Conv}_K(X))) \quad (8)$$

$$V = \text{SN}_V(\text{BN}_V(\text{Conv}_V(X))) \quad (9)$$

Here,  $Q$ ,  $K$ , and  $V \in \{0, 1\}^{T \times D \times H' \times W'}$ , retaining the same dimensions as the input sequence  $X$ . BN and Conv

represent batch normalization and two-dimensional convolution operations, respectively.  $\text{SN}(\cdot)$  means the spiking neuron layer.

The SDA mechanism consists of two variants: Spiking Single-Scale Dilated Self-Attention (SSDA) and Spiking Multi-Scale Dilated Self-Attention (SMDA). SSDA selects key and value matrices through the unfold operation with the corresponding dilation rate. It computes the single-scale self-attention along horizontal and vertical windows. SMDA achieves multi-scale information aggregation by performing SSDA across different heads. The SSDA of each head with unique dilation rate can capture features at different scales. By fusing the feature representations generated by these SSDA across different heads, SMDA effectively integrates multi-scale features at multiple levels. In our experiments, we adopt SMDA, where the  $Q$ ,  $K$ , and  $V$  matrices are split into three heads with respective dilation rates of 1, 2, and 3.

1) *Spiking Single-Scale Dilated Self-Attention*: For each spatial position of the query matrix, SSDA sparsely selects key and value matrices within horizontal and vertical windows to conduct self-attention. We define this process as:

$$S = \text{SSDA}(Q, K, V, r) \quad (10)$$

Here,  $Q$ ,  $K$ , and  $V$  are the spike-form query, key, and value matrices.  $r$  is the dilation rate that controls the sparsity of the sampling.  $S$  is the output after applying the SSDA operation.

Specifically, for the query matrix at spatial position  $(i, j)$ , SSDA sparsely selects  $K$  and  $V$  matrices within the horizontal window, defined as Eq. 11 and Eq. 12.

$$K_{\text{horizontal}} = [K(i, j + \delta_h \cdot r) \mid \delta_h \in \mathcal{R}_h] \quad (11)$$

$$V_{\text{horizontal}} = [V(i, j + \delta_h \cdot r) \mid \delta_h \in \mathcal{R}_h] \quad (12)$$

The sparse sampling process of  $K$  and  $V$  matrices along the vertical window are defined as Eq. 13 and Eq. 14.

$$K_{\text{vertical}} = [K(i + \delta_v \cdot r, j) \mid \delta_v \in \mathcal{R}_v] \quad (13)$$

$$V_{\text{vertical}} = [V(i + \delta_v \cdot r, j) \mid \delta_v \in \mathcal{R}_v] \quad (14)$$

Here,  $\delta_h$  and  $\delta_v$  denote the offset parameters for sampling positions in the horizontal and vertical windows relative to  $Q$ , respectively.  $r$  represents the dilation rate.  $\mathcal{R}_h$  and  $\mathcal{R}_v$  define the sampling ranges in the horizontal and vertical windows, as described in Eq. 15 and 16.

$$\mathcal{R}_h = \left\{ -\left\lfloor \frac{w_h}{2} \right\rfloor, -\left\lfloor \frac{w_h}{2} \right\rfloor + 1, \dots, \left\lfloor \frac{w_h}{2} \right\rfloor \right\} \quad (15)$$

$$\mathcal{R}_v = \left\{ -\left\lfloor \frac{w_v}{2} \right\rfloor, -\left\lfloor \frac{w_v}{2} \right\rfloor + 1, \dots, \left\lfloor \frac{w_v}{2} \right\rfloor \right\} \quad (16)$$

Here,  $\lfloor \cdot \rfloor$  denotes the floor operation.  $w_h$  and  $w_v$  are the sizes of the horizontal and vertical sliding windows, respectively. For instance, when the  $w_h$  and  $w_v$  are 3, both  $\mathcal{R}_h$  and  $\mathcal{R}_v$  are  $\{-1, 0, 1\}$ .

As shown in Fig. 3 (a), the self-attention operations of SSDA are independently computed within the horizontal and vertical windows. The process is described as Eq. 17.

$$\text{Attn} = (Q(K_{\text{horizontal}})^T V_{\text{horizontal}} * s^*) \odot (Q(K_{\text{vertical}})^T V_{\text{vertical}} * s^*) \quad (17)$$

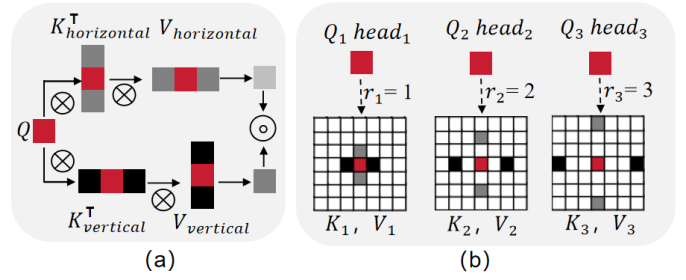


Fig. 3: The detailed implementations of SDA. (a) shows the computation process of attention in both horizontal and vertical windows. (b) illustrates that the query matrix slices sparsely select key and value matrices in the horizontal and vertical directions via different dilation rates ( $r_i = 1, 2$ , and  $3$ ).

Here,  $\text{Attn}$  represents the aggregate features of the horizontal and vertical.  $\odot$  denotes the concatenation operation along the channel dimension.  $s^*$  serves as scaling factor to constrain the large value of the matrix multiplication result, calculated as  $1/\sqrt{D}$ , a standard approach consistent with that proposed in [9], where  $D$  represents the channel dimension of  $Q$ .

2) *Spiking Multi-Scale Dilated Self-Attention*: To exploit the sparsity at different scales of the Spiking Dilated Self-Attention mechanism, we further propose a Spiking Multi-Scale Dilated Self-Attention to extract multi-scale semantic information. As shown in Fig. 2, given the  $Q$ ,  $K$ , and  $V$  matrices, and we partition the channels of these matrices to  $n$  different heads. As shown in Fig. 3 (b), each head is assigned a unique dilation rate to modulate the spatial spacing of the sampled key and value points, thereby enabling sampling across receptive fields of varying scales. Specifically, the  $i$ -th head of SMDA is formulated as follows:

$$M_i = \text{SSDA}(Q_i, K_i, V_i, r_i), \quad 1 \leq i \leq n \quad (18)$$

Here,  $r_i$  is the dilation rate of the  $i$ -th head, and  $Q_i$ ,  $K_i$  and  $V_i$  represent the slices of  $Q$ ,  $K$ , and  $V$  matrices fed into the  $i$ -th head, respectively.

After obtaining the multi-head outputs, they are concatenated along the channel dimension and then fed into a spiking neuron layer to generate spiking-form features. Finally, a convolution layer and a batch normalization (BN) layer are applied to aggregate the multi-scale features.

$$M = \text{SN}(\text{Conv}(\text{BN}(M_1 \odot M_2 \odot \dots \odot M_n))) \quad (19)$$

Here,  $\odot$  denotes the concatenation operation along the channel dimension, and  $M$  represents the final output of the SMDA.

#### IV. EXPERIMENTS

In this section, we first introduce the theoretical calculation methods for synaptic operation count and energy consumption (Section IV-A). Then we evaluate the classification performance of SDformer on static image datasets, including CIFAR-10 [29], CIFAR-100 [29]. We also explore the effectiveness of SDformer on neuromorphic datasets, including CIFAR10-DVS



TABLE I: Experimental results on CIFAR-10 and CIFAR-100. Note that "SDformer-4-384" means SDformer contains 4 encoder blocks and 384 feature embedding. "Spike" denotes "Spike-driven", "T" denotes "Timesteps", "Params" refers to "Parameters (M)", and "Accuracy" denotes "Top-1 Classification Accuracy(%)".

| Method                  | Architecture        | Spike | Params | T   | Accuracy     |              |
|-------------------------|---------------------|-------|--------|-----|--------------|--------------|
|                         |                     |       |        |     | CIFAR-10     | CIFAR-100    |
| Hybrid training [24]    | VGG-11              | ✗     | -      | 125 | 92.22        | 67.87        |
| STDP [36]               | CIFARNet            | ✗     | 17.54  | 12  | 89.83        | -            |
| STBP-tBN [37]           | ResNet-19           | ✗     | 12.63  | 4   | 92.92        | 70.86        |
| ResNet [38]             | ResNet-19           | ✗     | 12.63  | 1   | 94.97        | 75.35        |
| SEW-ResNet [39]         | SEW-ResNet-21/18    | ✗     | -      | 16  | 90.02        | 77.37        |
| Transformer [9]         | Transformer-4-384   | ✗     | 9.32   | 1   | 96.73        | 81.02        |
| Spikformer [17]         | Spikformer-4-384    | ✗     | 9.32   | 4   | 95.51        | 78.21        |
| Spikingformer [18]      | Spikingformer-4-384 | ✓     | 9.32   | 4   | 95.81        | 79.21        |
| S-Transformer [20]      | Spikformer-4-384    | ✓     | 10.28  | 4   | 95.60        | 78.40        |
| Spikingformer(CML) [19] | Spikingformer-4-384 | ✓     | 9.32   | 4   | 95.95        | 80.37        |
| SWformer [16]           | SWformer-4-384      | ✓     | -      | 4   | 96.10        | 79.30        |
| SAFormer [22]           | SAFormer-4-384      | ✓     | 9.32   | 4   | 95.80        | 79.07        |
| SDformer (Ours)         | SDformer-4-384      | ✓     | 9.32   | 4   | 96.27 ± 0.05 | 80.58 ± 0.06 |

[30] and DVS 128 Gesture [31] (Section IV-C). Our attention visualization analysis of SDformer proves its superiority (Section IV-D). Furthermore, we conduct ablation studies to assess the robustness and generalizability of the Spooling and SDA (Section IV-E). All experiments are conducted on a single server node with a 4090 GPU. The implementation is based on the leaky integrate-and-fire (LIF) neuron, PyTorch [32], SpikingJelly [33] and Pytorch image models library (Timm) [34]. We conduct three experimental trials and compute the mean values to avoid randomness. Our experimental results are reported as Top-1 accuracy (%) with mean ± standard deviation.

#### A. Theoretical Calculations of Synaptic Operation Count and Energy Consumption

To evaluate the theoretical energy consumption of SDformer, we first calculate the number of synaptic operations (SOPs):

$$\text{SOPs}(l) = fr \times T \times \text{FLOPs}(l) \quad (20)$$

Here,  $l$  is a block in SDformer,  $fr$  is the firing rate of the input spike sequence, and  $T$  is the spike neuron simulation timestep.  $\text{FLOPs}(l)$  refers to the floating-point operations of block  $l$ . Energy consumption is proportional to the number of operations. The theoretical energy consumption of SDformer is calculated:

$$E = 77\text{fJ} \times \text{SOPs} \quad (21)$$

Here, 77 fJ is the energy consumption per SOP [35].

#### B. Static Image Datasets

**- Implementation Details.** For a fair comparison with existing works, we followed the experimental setup of Spikformer [17], Spikingformer [18], and S-Transformer [20] to evaluate the performance of our SDformer. Our SDformer architecture consists of 4 encoder blocks with a feature embedding dimension of 384, and the patch size is  $4 \times 4$ . We employ the AdamW optimizer for 450 training epochs to ensure

sufficient learning and optimization time, including an initial 20 warm-up epochs and a final 10 cool-down epochs.

**- Results and Analysis.** As shown in Table I, On the CIFAR-10 dataset, SDformer achieves a top-1 classification accuracy of 96.27% with a theoretical energy consumption of only 1.23 mJ, significantly surpassing Spikformer (95.51%, 5.15 mJ) [17] by 0.76%. Comparing with SEW-ResNet (90.02%, 8.88 mJ) [39] further demonstrates the superiority of SDformer, showing a 6.25% increase in accuracy and an 86.14% reduction in energy consumption. Additionally, compared with S-Transformer (95.60%, 0.52 mJ) [20], SDformer delivers a substantial 0.67% improvement in top-1 classification accuracy, even with its slightly higher energy consumption. On the CIFAR-100 dataset, SDformer achieves a top-1 classification accuracy of 80.58% with a theoretical energy consumption of only 1.29 mJ, outperforming Spikformer [18] and SAFormer [22] by 2.37% and 1.51%, respectively. Specifically, compared to ResNet [38] and SEW-ResNet [39] methods employing ResNet architecture, SDformer exhibits improvements in accuracy by 5.23% and 3.21%, respectively.

#### C. Neuromorphic Datasets

**- Implementation Details.** For the neuromorphic Datasets, the SDformer with 2 encoder blocks, the patch embedding dimension is 256, and the patch size is  $16 \times 16$ . The timestep of the spiking neuron is 10 or 16. The AdamW optimizer is used, with 300 training epochs for CIFAR10-DVS, and 200 training epochs for DVS 128 Gesture. Each training process includes 10 warm-up epochs and 10 cool-down epochs.

**- Results and Analysis.** As shown in Table II, SDformer demonstrates competitive performance on neuromorphic datasets. On CIFAR10-DVS dataset, SDformer achieves a top-1 classification accuracy of 83.00% with 16 timesteps, significantly outperforming Spikformer [17], Spikingformer [18], and SAFormer [22] by 2.1%, 1.7%, and 1.7%, respectively. Even with only 10 timesteps, SDformer still achieves a top-1

TABLE II: Experimental results on CIFAR10-DVS and DVS128 Gesture. "Accuracy" denotes "Top-1 Classification Accuracy (%)", "T" denotes "Timesteps".

| Method             | CIFAR10-DVS |                  | DVS128 Gesture |                  |
|--------------------|-------------|------------------|----------------|------------------|
|                    | T           | Accuracy         | T              | Accuracy         |
| LIAF [40]          | 10          | 70.4             | 60             | 97.6             |
| tdBN [37]          | 10          | 67.8             | 40             | 96.9             |
| PLIF [41]          | 20          | 74.8             | 20             | 97.6             |
| Transformer [42]   | -           | 71.2             | -              | -                |
| Spikformer [17]    | 10          | 78.9             | 10             | 96.9             |
|                    | 16          | 80.9             | 16             | 98.3             |
| Spikingformer [18] | 10          | 79.9             | 10             | 96.2             |
|                    | 16          | 81.3             | 16             | 98.3             |
| SpikformerV2 [43]  | 10          | 78.9             | 10             | 96.9             |
|                    | 16          | 80.9             | 16             | 98.3             |
| TIM [21]           | 10          | 81.6             | -              | -                |
| SAFormer [21]      | 16          | 81.3             | 16             | 98.3             |
| SDformer (Ours)    | 10          | 81.60 $\pm$ 0.10 | 10             | 97.30 $\pm$ 0.15 |
| SDformer (Ours)    | 16          | 83.00 $\pm$ 0.15 | 16             | 98.60 $\pm$ 0.20 |

classification accuracy of 81.60%, improving 2.7% and 1.7% than Spikformer [17] and Spikingformer [18], respectively.

On DVS128 Gesture dataset, SDformer achieves a top-1 classification accuracy of 98.60% with 16 timesteps. The tdBN [37] achieves an accuracy of 96.90% with 40 timesteps. By contrast, SDformer not only improves top-1 classification accuracy by 1.70% but also reduces the number of timesteps by 60%. These results further validate the superior performance and effectiveness of our SDformer in neuromorphic classification tasks.

#### D. Attention Visualization

To further assess the classification performance of our proposed SDformer, we employ Grad-CAM [44] to visualize the attention maps generated by the last encoder block in Spikingformer [18] and our SDformer. As illustrated in Fig. 4, compared with Spikingformer, SDformer produces more continuous and concentrated attention regions, reflecting that SDformer exhibits a stronger ability to focus on task-relevant features. For instance, the shape of the ship (Figure 4 (a)) is horizontal, while that of the bird (Figure 4 (c)) is primarily vertical. SDformer effectively adapts to their directional distributions by allocating attention along the structural direction of objects. Moreover, SDformer demonstrates superior adaptability to scale variations within the same object category. For instance, the ship instances (Figures 4 (a) and (b)) differ substantially in spatial size. SDformer is capable of capturing both global information and fine-grained local details, indicating its robustness in multi-scale representation learning.

#### E. Ablation Study

We conduct ablation studies on the SPooling and SDA to analyze their specific impacts on model performance. Ablation experiments are performed on both static image datasets (CIFAR-10) and the neuromorphic datasets (CIFAR10-DVS).

**Spiking Response Pooling.** To validate the effectiveness of SPooling, we replace the max pooling in Spikingformer [18]

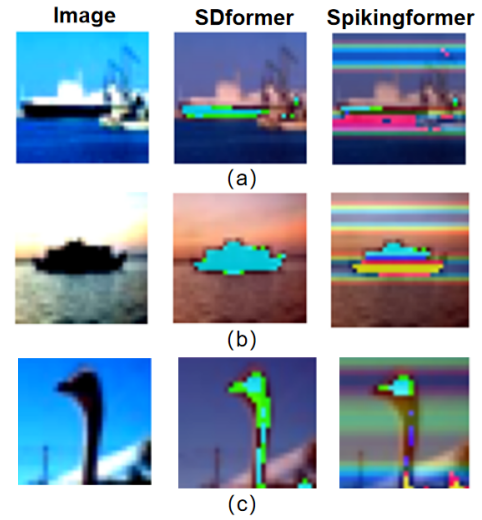


Fig. 4: Grad-CAM visualization of the last block of our SDformer and Spikingformer [18] on the CIFAR-10 dataset. (a)-(c) are demonstrations of 9 random samples extracted from the dataset.

(baseline) with SPooling, and maintain all other experimental settings unchanged. As shown in Table III, SPooling improves the top-1 classification accuracy of Spikingformer by 0.28% and 1.00% on CIFAR-10 and CIFAR10-DVS datasets, respectively. The experiments thoroughly demonstrate the superiority of SPooling compared with the traditional max pooling.

To further evaluate the specific contribution of SPooling in enhancing informative feature representation, we conduct a systematic analysis of the threshold parameter  $\lambda$  on the CIFAR-10 and CIFAR10-DVS datasets. As shown in Table IV, we have selected several representative threshold values of  $\lambda$  within the experimental range (covering both extreme and intermediate values). The best performance within the experimental range is achieved at  $\lambda = 0.4$  (96.27% on CIFAR-10 dataset and 83.00% on CIFAR10-DVS dataset). Notably, when  $\lambda = 0.0$ , the average pooling results in practice are typically non-negative and naturally meet or exceed the threshold  $\lambda$ . This means no average pooling results are filtered out, and SPooling degenerates into a weighted combination of max pooling and average pooling. In contrast, when  $\lambda = 1.0$ , most average pooling results are filtered out. As shown in Table IV, the performance degradation at these two extreme values ( $\lambda = 0.0$  and  $\lambda = 1.0$ ), highlights the essential role of  $\lambda$  as a balancing parameter, which facilitates more effective preservation of salient features. Except for the two extreme values exhibiting degraded performance, SPooling maintains consistently stable performance at all other values within the experimental range. This validates the effectiveness and robustness of our proposed SPooling.

**Spiking Dilated Self-Attention.** To verify the effectiveness of our proposed SDA, we replace the original SSA in Spikingformer with SDA. As shown in Table III, the variant model (with SDA) outperforms the baseline Spikingformer [18] (with

TABLE III: Ablation studies of SPooling and SDA on CIFAR-10 and CIFAR10-DVS datasets. The baseline is Spikingformer [18]. "Accuracy" denotes "Top-1 Classification Accuracy (%)".

| Model                    | Accuracy         |                  |
|--------------------------|------------------|------------------|
|                          | CIFAR-10         | CIFAR10-DVS      |
| Spikingformer            | 95.81            | 81.30            |
| Spikingformer + SPooling | 96.09            | 82.30            |
| Spikingformer + SDA      | 96.12            | 82.50            |
| SDformer (Ours)          | 96.27 $\pm$ 0.05 | 83.00 $\pm$ 0.15 |

TABLE IV: Impact of different threshold parameters  $\lambda$  in SPooling on CIFAR-10 and CIFAR10-DVS datasets. "Accuracy" denotes "Top-1 Classification Accuracy (%)".

| $\lambda$  | Accuracy                           |                                    |
|------------|------------------------------------|------------------------------------|
|            | CIFAR-10                           | CIFAR10-DVS                        |
| 0.0        | 95.82 $\pm$ 0.03                   | 81.90 $\pm$ 0.20                   |
| 0.2        | 95.56 $\pm$ 0.02                   | 81.47 $\pm$ 0.15                   |
| 0.3        | 95.94 $\pm$ 0.04                   | 82.43 $\pm$ 0.21                   |
| <b>0.4</b> | <b>96.27 <math>\pm</math> 0.05</b> | <b>83.00 <math>\pm</math> 0.15</b> |
| 0.5        | 96.06 $\pm$ 0.06                   | 81.73 $\pm$ 0.15                   |
| 1.0        | 95.84 $\pm$ 0.04                   | 81.83 $\pm$ 0.31                   |

SSA) by 0.31% and 1.20% in top-1 classification accuracy on CIFAR-10 and CIFAR10-DVS datasets, respectively. These results clearly validate the effectiveness and superiority of our proposed SDA, confirming that replacing the original SSA with SDA can significantly enhance the feature extraction capability of model and improve classification performance.

**The Influence of Dilation Parameters.** We analyze the impact of different dilation scales in SDA on classification performance using the CIFAR-10 and CIFAR10-DVS datasets. As shown in Table V, we have selected several representative dilation rate combinations (covering both SSDA and SMDA), to focus on analyzing their effects on classification performance. SDA demonstrates robustness to variations in dilation rates, with only slight variations in accuracy across different dilation configurations.

Spiking Multi-Scale Dilated Self-Attention, particularly with dilation rates [1, 2, 3], statistically significantly outperforms all Spiking Single-Scale Dilated Self-Attention configurations (i.e., [1], [2], and [3]) with  $p < 0.01$  on both datasets. This indicates that multi dilated-scale can provide more comprehensive information than single dilated-scale. When a dilation rate of 4 is included (e.g., [2,3,4], [1,2,3,4]), the top-1 classification accuracy on both CIFAR-10 and CIFAR10-DVS slightly decreases. This is because an excessively large dilation rate (e.g., 4) creates overly sparse sampling in local windows, breaking the continuity of meaningful local features and introducing irrelevant spatial noise. Such sparsity weakens the ability of model to learn discriminative spatial relationships, directly leading to performance degradation. These results suggest that Spiking Dilated Self-Attention achieves its best trade-off between performance and efficiency with multi-scale dilation without excessively large dilation rates.

TABLE V: Comparative impact of dilation parameters on classification accuracy for CIFAR-10 and CIFAR10-DVS datasets. Note that "Accuracy" denotes "Top-1 Classification Accuracy (%)". Statistical significance (independent samples t-test) is marked as: \* $p < 0.05$ , \*\* $p < 0.01$ , compared to the "SMDA-[1,2,3]".

| Scale | Dilation Rate | Accuracy           |                    |
|-------|---------------|--------------------|--------------------|
|       |               | CIFAR-10           | CIFAR10-DVS        |
| SSDA  | [1]           | 95.86 $\pm$ 0.06** | 81.87 $\pm$ 0.15** |
|       | [2]           | 95.56 $\pm$ 0.08** | 81.77 $\pm$ 0.15** |
|       | [3]           | 95.54 $\pm$ 0.06** | 81.30 $\pm$ 0.20** |
| SMDA  | [1,2,3]       | 96.27 $\pm$ 0.05   | 83.00 $\pm$ 0.15   |
|       | [2,3,4]       | 96.00 $\pm$ 0.14*  | 82.53 $\pm$ 0.25** |
|       | [1,2,3,4]     | 95.92 $\pm$ 0.18** | 81.97 $\pm$ 0.21** |

## V. CONCLUSION

In this paper, we propose a novel Spiking Dilated Self-Attention mechanism, which computes self-attention within the local dilated windows along horizontal and vertical direction, effectively capturing multi-scale features. Furthermore, we introduce Spiking Response Pooling (SPooling), designed to extract richer and more informative feature representations. Extensive experiments on both static and neuromorphic datasets demonstrate that our approach consistently achieves state-of-the-art performance. Notably, we pioneer the validation of local dilated window attention in directly trained spiking transformers. In the future, we plan to explore specific hardware optimizations and conduct benchmarking on real platforms to comprehensively assess the practical potential of our methods in real-world deployment scenarios.

## VI. ACKNOWLEDGMENT

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